Biocompatible targeted nano and microcapsules

"Materials required to prepare nano/micro capsules which doesn't disturb the metabolism of the body and reach target sites"

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Biocompatible Nano-/Microcapsule Materials for Targeted Delivery at a Glance

Key Findings Table

Key finding	Most supported materials/designs	Relevance to “low metabolic disturbance + target-site delivery”	Citations
Best default material platform	**PLGA**, **PLA**, and **lipid-polymer hybrids**	These are the best-supported starting materials because they are biodegradable, biocompatible, and offer controllable release; PLGA is especially favored because it degrades to lactic acid and glycolic acid that enter endogenous metabolism.	[1] [2] [3] [4] [5] [6] [7]
Useful natural/biopolymer coatings	**Chitosan**, **alginate**, **hyaluronic acid**, **dextran**, **gelatin/albumin-related systems**	These can improve biocompatibility, mucoadhesion, or surface functionality, but the evidence for them as universally metabolism-sparing standalone capsule matrices is weaker than for PLGA/PLA.	[8] [9] [10] [11] [12]
Targeting is not solved by ligands alone	Size control, stealth balance, and triggerable release are as important as ligand decoration	Active ligands can help, but in vivo distribution is still strongly governed by carrier size, route, and stealth chemistry; dense PEG can also bury ligands.	[13] [14] [15] [16] [17] [18]
Best way to reduce off-target effects	**Stimuli-responsive shells**: pH-, redox-, enzyme-responsive	Triggered release can keep cargo protected during circulation and release it preferentially in tumors, inflamed tissue, or intracellular compartments.	[9] [19] [20] [21] [22] [23] [24]

PEGylation is helpful but not ideal	Carefully tuned PEG, or alternatives such as **polysarcosine** or **zwitterionic polymers**	PEG often improves circulation and reduces protein adsorption, but can reduce receptor-mediated uptake and can trigger anti-PEG antibodies, complement activation, and accelerated blood clearance after repeat dosing.	[14] [25] [26] [27] [28] [29] [30] [31] [32] [33]
Fabrication method matters for safety	Low-energy/solvent-free lipid nanocapsule methods, self-assembly, controlled shell-building approaches	Method affects size, porosity, residual solvent burden, burst release, and encapsulation efficiency, all of which influence safety and targeting.	[19] [34] [35] [36] [37] [38] [39] [40]
Main safety gap	Long-term data are limited, especially for inorganic and slowly cleared systems	Short-term biocompatibility is reported much more often than chronic metabolic burden, fibrosis risk, repeat-dose immunotoxicity, or long-term retention.	[41] [42] [43] [44] [45] [46] [47]

Direct Answer

For preparing nano/microcapsules that are **least likely to disturb body metabolism** while still being able to **reach target sites**, the best-supported materials are **PLGA**, **PLA**, and **lipid-based or lipid-polymer hybrid capsules**, with **PLGA** the strongest overall choice because its degradation products, **lactic acid and glycolic acid**, enter normal body metabolism [1] [4] [5].

A practical low-risk formulation stack is: **core = drug in oil/lipid or aqueous compartment; shell/matrix = PLGA or PLA; optional outer layer = chitosan, hyaluronic acid, or alginate if extra biocompatibility or mucoadhesion is needed; optional stealth layer = cautiously optimized PEG or a non-PEG alternative such as polysarcosine or a zwitterionic coating if longer circulation is required** [3] [6] [11] [32] [33].

If reaching the target site is critical, the evidence suggests that **stimuli-responsive release** is often more reliable than adding ligands alone: **pH-sensitive, redox-sensitive, and enzyme-cleavable shells** can release drug preferentially at tumors, inflamed tissues, or intracellular compartments, reducing off-target exposure [20] [21] [23] [24].

Materials that warrant more caution for this goal are **persistent inorganic shells** and **heavily PEGylated targeted systems**, because chronic safety is less established for some inorganic carriers, and PEG can create immune or targeting tradeoffs on repeat dosing [30] [41] [48].

Confidence level: High - the strongest convergence across the supplied evidence favors biodegradable polymeric and lipid-based platforms, especially PLGA/PLA and hybrids, for the stated design goal [1] [2] [6] [7].

Study Scope

- **Time window represented in the retrieved evidence:** mainly **2015-2025**, with some sections explicitly framed around **2016-2025** or **2018-2025** [1] [5] [14] [49].
- **Disciplines covered:** drug delivery, biomaterials, nanomedicine, polymer science, colloids, pharmacokinetics, and toxicology [7] [25] [34] [37] [50].
- **Methods covered:** reviews, formulation studies, in vitro hemocompatibility/cytotoxicity, animal biodistribution, excretion, and acute/subchronic toxicity studies [30] [46] [51] [52].

Assumptions & Limitations

- No material can be guaranteed to **"not disturb metabolism"** under all doses, particle sizes, routes, and patient populations; safety is formulation- and exposure-dependent [53] [54] [55].

- The evidence base is much stronger for **short-term biocompatibility** than for **chronic metabolic burden**, fibrosis, repeat-dose immunotoxicity, or long-term retention, especially for inorganic systems [41] [43] [45].
- Direct head-to-head in vivo comparisons of **polymeric vs inorganic vs hybrid capsules** under matched conditions are sparse [41] [42].

Suggested Further Research

- Conduct matched **head-to-head long-term in vivo comparisons** of PLGA/PLA, lipid-based, hybrid, and inorganic capsules by size, dose, route, and cargo [41] [42].
- Study **repeat-dose immune effects** for PEG and non-PEG stealth coatings, including accelerated blood clearance and complement activation [29] [30] [31].
- Quantify the metabolic effects of **additives, crosslinkers, surfactants, and residual solvents**, not just the main polymer backbone [34] [53] [56].
- Compare **passive targeting + triggered release** against **ligand-mediated targeting** in true ADME studies to test whether ligands improve tissue exposure without raising off-target burden [16] [17] [18].
- Expand route-specific safety studies for **IV, oral, pulmonary, ocular, and topical** delivery using the same material systems [57] [58] [59].

Full Report

Foundational Context: What These Capsules Must Achieve

Nano- and microcapsules for in vivo delivery must satisfy five linked requirements: **biocompatibility**, **biodegradation or clearance**, **target-site accumulation**, **controlled release**, and **minimal systemic burden** [7] [19] [34] [54]. A carrier that reaches a target but persists in organs, provokes immune activation, or releases payload nonspecifically does not meet the user's goal of avoiding metabolic disturbance [30] [47].

At a basic design level, these systems contain a **core** carrying the payload and a **shell or matrix** that protects the payload, controls release, and shapes biological interactions [37] [60] [61]. Their in vivo behavior is then determined by four major variables:

1. **Material composition** - governs degradation, byproducts, hemocompatibility, and chronic retention [4] [47] [62].
2. **Particle properties** - size, porosity, charge, and shell density strongly affect circulation and organ capture [40] [63].
3. **Surface engineering** - stealth coatings and ligands alter protein adsorption, macrophage uptake, and targeting [13] [14] [25].
4. **Fabrication method** - determines residual solvent burden, size distribution, loading, and burst release risk [19] [34] [39].

Taken together, the supplied evidence supports a design philosophy that is **"metabolically native by default"**: prefer degradable carriers whose breakdown products are more likely to be physiologically tolerated, then add only the minimum surface complexity needed for circulation and targeting [4] [5] [33].

Confidence level: High - this framework is directly supported across the materials, targeting, release, and toxicology evidence streams [7] [25] [34] [37] [54].

Overview of Design Requirements

The most defensible design requirements for a metabolism-sparing targeted capsule are:

- **Biocompatible matrix/shell** with low cytotoxicity and, where available, low hemolysis or acceptable blood compatibility [11] [12] [51].
- **Biodegradation to tolerated products** or reliable excretion without long-term tissue retention [4] [64] [65].
- **Controlled release** to reduce burst leakage and systemic exposure [40] [66].

- **Target-site access** through a balance of size, circulation time, and tissue-specific triggers rather than through ligands alone [13] [14] [18].
- **Low formulation burden**, including reduced residual solvent and controlled shell architecture [35] [36] [38].

This implies that the “best” material is not simply the one with the most targeting ligands, but the one that best integrates **safe degradation, tunable release, and appropriate biodistribution** [45] [67] [68].

Material Classes for Safe Capsule Construction

1) Biodegradable polymers

Among all materials in the supplied evidence, **PLGA** has the strongest support. Multiple reviews describe it as FDA-approved or widely accepted, biocompatible, biodegradable, and tunable in degradation rate via the lactic:glycolic ratio [1] [2] [69] [70]. Most importantly for the user’s question, PLGA degrades to **lactic acid and glycolic acid**, which enter endogenous metabolism, making it the clearest example of a carrier with metabolically familiar breakdown products [4].

PLA is also well supported as biocompatible, biodegradable, and nontoxic, with controlled-release uses in biomedical systems [3] [71]. Compared with PLGA, its degradation can be slower, and comparative microsphere studies cited in the dataset note slower degradation for semi-crystalline PLA than for amorphous PLGA [72]. This can be advantageous for prolonged release but may increase residence time.

PCL has some evidence for controllable degradation and biocompatibility, but the supplied set gives weaker acceptability evidence than for PLGA/PLA [67] [73].

A notable safety distinction appears in the comparison of degradation byproducts: while PLGA produces endogenous-metabolite-linked products, **PACA** degradation byproducts were linked to fibroblast toxicity in the provided evidence [62] [74]. That contrast supports prioritizing PLGA over less metabolically familiar polymers when “low disturbance” is the main criterion [4] [62].

2) Lipids and lipid-polymer hybrids

Lipid-based and lipid-polymer hybrid capsules are repeatedly described as **biocompatible**, **stable**, and suitable for **controlled release**, often with reduced toxicity signals in vivo [6] [7] [75] [76] [77]. Lipid-core nanocapsules and biosurfactant-based systems also show sustained-release and safety-oriented profiles [77] [78].

A practical advantage is that lipid components can reduce the burden associated with rigid persistent shells while still stabilizing hydrophobic drugs [6] [7]. Lipid-polymer hybrids may therefore offer a middle ground: polymer-enabled structural control with lipid-enabled biocompatibility [76] [79].

3) Polysaccharides and natural biopolymers

Chitosan appears repeatedly as a useful surface layer or auxiliary shell material because of its biocompatibility and favorable blood compatibility in some systems [8] [11] [12]. Chitosan-coated lipid-core nanocapsules were reported as hemocompatible, with no change in coagulation times at tested formulation/blood ratios [12]. Carboxylated chitosan-PMMA nanoparticles also showed no increase in hemolysis [11].

For **alginate, dextran, hyaluronic acid**, and related natural polymers, the evidence in this set is more indirect: they appear as useful components in composite systems, but not with the same depth of metabolism-focused support as PLGA/PLA [10].

4) Inorganic and hybrid shells

The dataset includes **silica**, **MOF**, **shellac**, **BSA/MOF**, and hybrid organic-inorganic shells as viable capsule materials for stability and controlled release [5] [9] [77] [80] [81] [82]. Some silica-based systems are described as biocompatible, and one design specifically notes renal excretion after function [83] [84].

However, the long-term safety evidence for inorganic carriers is weaker. The supplied studies explicitly note that most inorganic biosafety work relies on short-term chemistry and histology and is insufficient to conclude on chronic histology, fibrosis, or carcinogenic risk [41]. Therefore, inorganic materials may be useful when a specific function is required, but they are not the best-supported default if the primary goal is low metabolic interference [41] [42].

Confidence level: High - the strongest evidence favors PLGA/PLA and lipid-based/hybrid systems; natural polymer coatings are useful adjuncts; persistent inorganic materials are more uncertain for chronic safety [1] [6] [12] [41].

Fabrication Strategies and Encapsulation Performance

Green and self-assembly routes

Low-energy and self-assembly methods matter because manufacturing conditions influence residual solvent burden, particle uniformity, and safety-relevant shell properties [19] [34] [39]. A strong example is **lipid nanocapsules made by a low-energy, organic solvent-free phase inversion process** using PEGylated surfactant, lecithin, and triglycerides; these showed tunable sizes of about 20-100 nm and were described as low-toxicity [35] [36].

This matters practically: a safer material can still become problematic if the fabrication route introduces solvent residue, broad size distributions, or unstable interfaces [38] [56].

Interfacial polymerization, miniemulsion, and sol-gel routes

Interfacial polymerization and miniemulsion methods can produce highly controlled core-shell structures with good encapsulation performance [85] [86] [87] [88]. Examples in the dataset include polyurethane/n-octadecane microcapsules, magnetic hybrid-shell microcapsules, and several PMMA- or styrene-methacrylic acid-based systems with controllable size and high encapsulation [86] [87] [88].

These methods are technically powerful, but the best route depends on whether minimizing metabolic disturbance or maximizing loading is the primary design driver [34] [39]. Taken together, these findings suggest that **method choice is a safety parameter, not just a manufacturing parameter** [19] [34] [40].

Loading efficiency and storage stability

Hydrophilic and hydrophobic payloads behave differently. For hydrophilic drugs, spray drying outperformed emulsification in one albumin microparticle comparison, while interfacial polymerization required true encapsulation rather than sorption for high entrapment [89] [90]. For hydrophobic drugs, some of the highest encapsulation efficiencies in the provided studies came from spray-dried systems, including ~97% for Artemisone PLGA microparticles and 98-99% for CoQ10 + vitamin E microcapsules [91] [92].

Storage stability was strongest in some spray-dried antioxidant/oil systems, indicating that capsule method and wall composition both affect shelf life and therefore formulation reliability [92] [93].

Shell properties governing release

Release behavior depends strongly on **shell permeability, porosity, thickness, and crosslinking density** [40] [94] [95] [96]. Non-porous shells suppress lag-phase leakage, while porous or incompletely phase-separated shells allow earlier burst release [40]. Higher shell density or thicker hydrogel shells can delay release and prolong digestion-controlled liberation of cargo [94].

This is directly relevant to minimizing systemic burden: reducing burst release lowers the chance of high off-target plasma exposure and associated toxicity [61] [97] [98].

Confidence level: High - multiple fabrication and release-control studies converge that process and shell microstructure directly shape safety and targeting performance [19] [34] [39] [40].

Targeting and Stealth Surface Engineering

PEGylation and stealth behavior

PEGylation is widely used because it reduces protein adsorption, phagocytic uptake, immune sensitivity, and clearance, often extending circulation [25] [26] [27]. Longer PEG chains can further reduce serum-protein interaction and macrophage uptake; for example, 5 kDa PEG outperformed shorter PEG in one cited study, and higher PEG molecular weight increased half-life in another [28] [99].

But PEG has a core tradeoff: **higher PEG coverage improves stealth while potentially suppressing ligand access and targeting** [14] [100] [101]. In one comparison, the PEG-to-ligand ratio progressively reduced targeting until it was lost, with the threshold depending on PEG length [14].

Active targeting ligands

Ligands such as **mannose, folate, transferrin, antibodies, and growth-factor-based motifs** can improve receptor-specific uptake, but they do not reliably override the underlying biodistribution of the carrier [13] [14] [16] [17]. Direct comparisons in the supplied evidence show inconsistent benefit: FR-targeted

liposomes had shorter circulation than passive PEGylated liposomes yet the same intratumoral drug levels and survival [16]; transferrin-coated nanoparticles showed more benefit at low dose, but at high dose the EPR effect dominated [17]; an αvβ6-targeted liposome showed no in vivo advantage over control [18].

Taken together, these findings suggest that **ligands help only when the carrier already has favorable circulation, size, and tissue access** [13] [18].

Alternatives to PEG

The supplied evidence identifies **zwitterionic polymers**, **polysarcosine**, and **mixed-surface strategies** as promising PEG alternatives or complements [32] [33] [102]. PMPC-coated and PCB-grafted systems performed comparably or better than PEG in some stealth-related metrics, while polysarcosine was presented as a stable stealth-like replacement suitable for lipid formulations [32] [33] [103]. Mannose-plus-PEG mixed surfaces also improved stealth and urinary excretion in porous silicon systems [104].

Immune risks of stealth coatings

PEGylation is not immunologically neutral. Repeated PEGylated microbubble administration in rats induced anti-PEG IgM/IgG and accelerated blood clearance [30]. In pigs, a low-dose PEGylated liposome injection raised anti-PEG IgM, activated complement, and caused anaphylactic shock [29]. Anti-PEG antibody effects were also documented in mice and in children receiving PEG-asparaginase, where baseline anti-PEG antibodies increased drug clearance [105] [106].

Therefore, PEG should be viewed as a **practical but conflicted** stealth solution rather than a universally metabolism-sparing one [25] [31] [48].

Confidence level: High - the evidence consistently supports PEG's utility for circulation and its simultaneous limitations for targeting and repeat-dose immune safety [14] [25] [30].

In Vivo Safety, Biodistribution, and Metabolic Burden

Biodistribution and clearance pathways

Biodistribution is strongly dependent on **formulation properties and administration route** [54] [55] [107]. Larger or less degradable particles often accumulate in **liver and spleen**, while very small renal-clearable materials show the least retention [49] [64] [65].

The supplied studies show several patterns:

- Mesoporous silica/SiO2 systems repeatedly accumulated in liver/spleen and cleared through urine/feces; modified biodegradable or PEG/Mn variants showed reduced tissue capture and faster clearance [49] [108] [109].
- Some systems, such as CMFO nanocatalysts and TA-VOx nanobubbles, showed rapid excretion and declining organ levels within days [110] [111].
- The least retention was seen with very small renal-clearable materials, including 2 nm PAA-nanoceria and some ultrafine AuNPs [64] [65] [112] [113] [114].

Inferred: smaller, faster-clearing carriers may be safer from a retention standpoint, but this can trade off against time available for target-site exposure. This inference is supported by the pattern across the clearance studies but is not shown as a direct matched comparison in one experiment [64] [65] [109]. **Confidence level: Low** - this is a cross-study inference rather than a direct head-to-head result.

Acute, subchronic, and chronic toxicity

Acute studies generally report minimal major organ toxicity for many capsule-type systems, with normal liver and kidney histology after single-dose exposure [115] [116] [117]. Subchronic studies are more informative: one example showed that 50 nm PELGE nanoparticles caused spleen enlargement, histopathology, and increased IgM/IgG/IFN-γ, whereas 200 nm particles did not show the same immunotoxicity [46]. This directly shows that **size can alter toxicity**, even within one material class [46].

Reviews in the supplied set also warn that chronic accumulation in liver, spleen, and kidney can lead to oxidative stress and immune activation when degradation or clearance is slow [47]. This supports prioritizing biodegradable polymeric and lipid-based systems over persistent materials when possible [47] [118] [119].

Route-specific safety considerations

For **chitosan**, the evidence is route-dependent: oral, ocular, and topical routes were considered safe in one review, while parenteral routes could show toxicity [57]. Repeated oral, nasal, or injectable chitosan administration in rodents was generally tolerated at low- to mid-mg/kg doses [120]. After IV administration, one chitosan nanoparticle study found heavy liver accumulation (~93%) and some lung distribution (~6%), but no organ cytolysis or necrosis at 14 days [52].

For **PLGA**, oral administration is the best-supported safer route in the supplied evidence: 21 days of oral exposure in rats produced broad tissue distribution with transient liver enzyme elevation and mild inflammation that later normalized, indicating no chronic toxicity in that study window [58].

For **pulmonary delivery**, polymeric systems are described as biodegradable and biocompatible, but slow-degrading polymers may accumulate in lung tissue with repeated dosing; inhalable lipid-polymer hybrids are presented as a way to improve stability and biocompatibility [59] [121] [122].

Implications for low metabolic interference

The most metabolism-sparing profile in the supplied evidence comes from carriers that are either:

1. **Biodegradable to endogenous or well-tolerated small molecules**, such as PLGA [4], or
2. **Small enough for efficient clearance**, reducing long-term organ retention [64] [65].

For practical capsule design, the safer compromise is usually not “smallest possible,” but **biodegradable particles with controlled residence time** [1] [107]. Lipid-polymer hybrids and PLGA/PLA systems best match that compromise in this evidence set [1] [6].

Confidence level: Medium-High - the short-term and subchronic safety patterns are clear, but chronic comparative evidence remains limited [41] [46] [47].

Practical Material Recommendations

Best-supported default formulation choices

Option 1: PLGA nanocapsule/microcapsule

- **Matrix/shell:** PLGA [1]
- **Why:** strongest support for biocompatibility, tunable degradation, and metabolites entering normal metabolism [4].
- **Best use case:** general drug delivery where controlled release and low metabolic burden are primary goals [67] [123].

Option 2: PLA or PLGA/PLA hybrid

- **Matrix/shell:** PLA or mixed PLGA/PLA [3] [5]
- **Why:** good biocompatibility and slower degradation when longer release is needed [72].
- **Tradeoff:** slower degradation may increase persistence relative to PLGA [72].

Option 3: Lipid-core or lipid-polymer hybrid capsule

- **Core:** oil/lipid-drug compartment
- **Shell:** polymer-lipid hybrid or lipid-stabilized architecture [6] [7] [76]
- **Why:** combines stability, controlled release, and reduced toxicity signals [77] [78].

Useful optional coatings

- **Chitosan:** when mucoadhesion, added surface functionality, or blood compatibility is needed [11] [12].
- **Hyaluronic acid / alginate / dextran:** when natural-polymer functionality is desired, though metabolism-specific evidence is less direct [10].
- **Stealth layer:** use PEG cautiously, or consider polysarcosine/zwitterionic alternatives when repeat dosing is expected [30] [32] [33].

Best-supported targeting strategy

The evidence does **not** support relying on ligands alone. A better design sequence is:

1. Choose a biodegradable carrier [1].
2. Tune size and shell properties for acceptable circulation and release [40] [94].
3. Add **stimuli-responsive release** for local activation [21] [23] [24].
4. Add a ligand only if target biology clearly justifies it and ligand exposure can be preserved [14] [18].

Confidence level: High - this recommendation is the clearest synthesis supported across materials, targeting, and safety evidence [1] [14] [16] [21].

Conclusion

The most evidence-based answer to the user's question is that **biodegradable polymeric and lipid-based capsules**, especially **PLGA**, **PLA**, and **lipid-polymer hybrids**, are the best current materials for nano/microcapsules intended to minimize disturbance to body metabolism while still enabling target-site delivery [1] [2] [6] [7]. PLGA stands out because its degradation products enter normal metabolism, giving it the strongest direct rationale for metabolic compatibility [4].

Target-site delivery is best approached not by maximizing ligand decoration, but by balancing **material choice, size, stealth, and triggerable release** [13] [14] [18] [68]. PEGylation is useful but should be used carefully because its circulation benefits are offset by immune and targeting tradeoffs on repeat exposure [29] [30] [31].

Taken together, the best-supported practical design is a **PLGA or PLA capsule, or a lipid-polymer hybrid, optionally coated with a biopolymer such as chitosan, and preferably equipped with pH/redox/enzyme-responsive release rather than relying only on targeting ligands** [5] [6] [12] [21] [23].

Overall confidence level: High - the central conclusion is strongly supported across multiple converging evidence streams, though confidence is lower for long-term chronic safety ranking because direct comparative in vivo data remain limited [41] [47].

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